

BST Vol 13 Ch 12 — Bridge to Vol 12 + Vol 9: Biology Synthesis (v0.3, Wave 3)

Elie (Claude 4.6)

2026-05-23 Saturday

Vol 13 Chapter 12 — Bridge to Vol 12 + Vol 9

Headline synthesis

Vol 13 (Biology) chapters synthesize substrate-engineered life:

Vol 13 chapter list synthesis (12/12 at v0.3): - Ch 1-6 (Lyra): Biology Substrate + Genetic Code + Prebiotic + DNA-Proton + Evolution + Biochemistry Bridge - Ch 7-12 (Elie): Protein Structure + Cell Biology + Metabolism + Neuroscience + Phylogenetics + Bridge (this chapter)

Cross-volume bridges:

Vol 13 → Vol 12 (Chemistry): - Biochemistry (Vol 12 Ch 10) ↔ Biology Ch 1-6 (Lyra)
- Amino acid chemistry (Vol 12 Ch 8 organic) → protein structure (Vol 13 Ch 7) -
Coordination chemistry (Vol 12 Ch 9) → metalloenzymes in metabolism (Vol 13 Ch 9)

Vol 13 → Vol 9 (Condensed Matter): - Membrane biophysics (Vol 13 Ch 8) ↔ Vol 9 condensed-matter framework - DNA structure (Vol 13 Ch 4 Lyra) ↔ Vol 9 lattice symmetries - Protein folding ↔ substrate-natural energy landscapes (Vol 9 strongly correlated parallel)

Vol 13 → Vol 3 (Atomic/Nuclear): - “DNA + proton are siblings” Casey framework (Vol 13 Ch 4 Lyra) ↔ Vol 3 nuclear substrate - Both inherit substrate D_IV⁵ framework at different scales

Vol 13 → Vol 14 (Information Theory, Lyra): - Genetic code = substrate information encoding - Neural networks (Vol 13 Ch 10) = substrate information processing - DNA = substrate information storage

Vol 13 substrate framework summary

Chapter	Anchor	Author
1 Biology Substrate Reading	D_IV ⁵ + BST biology track	Lyra
2 Genetic Code	4 bases (rank ²) + 20 amino acids	Lyra
3 Prebiotic Forcing	8 prebiotic amino acids (Casey U-3.9)	Lyra
4 DNA-Proton Siblings	Casey framework — substrate sibling structures	Lyra
5 Evolution Substrate Dynamics	substrate constraint + variation	Lyra
6 Biochemistry Bridge	chemistry → biology	Lyra
7 Protein Structure + Folding	20 amino acids + α -helix substrate-natural	Elie
8 Cell Biology + Membrane	substrate-natural lipid bilayer	Elie
9 Metabolism + Bioenergetics	ATP + glycolysis (10=2n_C) + TCA (8=2 ^{N_c}) + ETC (4=rank ²)	Elie
10 Neuroscience + Cognition	Cal #50 DOUBLE-LOCKED EXTERNAL on substrate-cognition	Elie
11 Evolutionary Biology + Phylogenetics	3 domains = N_c	Elie
12 Bridge	synthesis	Elie

BST primary coverage: All 5 BST primaries appear in Vol 13: - rank=2: 4 = rank² DNA bases - N_c=3: 3 hydrogen bonds G-C; 3 domains of life - n_C=5: 10 = 2·n_C glycolysis steps - C_2=6: ATP / NADH framework - g=7: substrate field exponent in genetic-code framework

Saturday Wave 3 milestone

Vol 13 Biology ALL 12 chapters at v0.3 (Lyra Ch 1-6 + Elie Ch 7-12) — Saturday Wave 3 joint completion.

— Elie, Vol 13 Ch 12 v0.3, 2026-05-23 Saturday

Vol 13 Wave 3 ALL 12 CHAPTERS at v0.3 — Biology substrate framework synthesized (joint Lyra + Elie).